regress — Linear regression+

+This command includes features that are part of StataNow.

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Description

`regress` performs ordinary least-squares linear regression. `regress` can also perform weighted estimation, compute robust and cluster–robust standard errors, and adjust results for complex survey designs.

Quick start

Simple linear regression of `y` on `x1`

```
regress y x1
```

Regression of `y` on `x1`, `x2`, and indicators for categorical variable `a`

```
regress y x1 x2 i.a
```

Add the interaction between continuous variable `x2` and `a`

```
regress y x1 c.x2##i.a
```

Fit model for observations where `v1` is greater than zero

```
regress y x1 x2 i.a if v1>0
```

With cluster–robust standard errors for clustering by levels of `cvar`

```
regress y x1 x2 i.a, vce(cluster cvar)
```

With cluster–robust standard errors for clustering by levels of `cvar1` and `cvar2`

```
regress y x1 x2 i.a, vce(cluster cvar1 cvar2)
```

With Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors (StataNow)

```
regress y x1 x2 i.a, vce(hac nwest)
```

With bootstrap standard errors

```
regress y x1 x2 i.a, vce(bootstrap)
```

Report standardized coefficients

```
regress y x1 x2 i.a, beta
```

Adjust for complex survey design using `svyset` data

```
svy: regress y x1 x2 i.a
```

Use sampling weight `wvar`

```
regress y x1 x2 i.a [pweight=wvar]
```

Menu

Statistics > Linear models and related > Linear regression

Syntax

```
regress depvar [indepvars] [if] [in] [weight] [, options]
```

options	Description
Model	
noconstant	suppress constant term
hascons	has user-supplied constant
tsscons	compute total sum of squares with constant; seldom used
SE/Robust	
vce(*vcetype*)	*vcetype* may be ols, robust, cluster *clustvarlist*, bootstrap, jackknife, hc2 [*clustvar*], or hc3
+vce(*vcetype*)	*vcetype* may also be hc3 [*clustvar*] or hac *hacspec*
Reporting	
level(#)	set confidence level; default is level(95)
beta	report standardized beta coefficients
eform(*string*)	report exponentiated coefficients and label as *string*
depname(*varname*)	substitute dependent variable name; programmer's option
clustertable	display table of multiway cluster combinations
display_options	control columns and column formats, row spacing, line width, display of omitted variables and base and empty cells, and factor-variable labeling
noheader	suppress output header
notable	suppress coefficient table
plus	make table extendable
mse1	force mean squared error to 1
coeflegend	display legend instead of statistics

+This feature is part of StataNow.

indepvars may contain factor variables; see **[U] 11.4.3 Factor variables**.

depvar and *indepvars* may contain time-series operators; see **[U] 11.4.4 Time-series varlists**.

bayes, bayesboot, bootstrap, by, collect, fmm, fp, jackknife, mfp, mi estimate, nestreg, rolling, statsby, stepwise, and svy are allowed; see **[U] 11.1.10 Prefix commands**. For more details, see [BAYES] **bayes: regress** and [FMM] **fmm: regress**.

vce(bootstrap) and vce(jackknife) are not allowed with the mi estimate prefix; see [MI] **mi estimate**.

Weights are not allowed with the bootstrap prefix; see [R] **bootstrap**.

aweights are not allowed with the jackknife prefix; see [R] **jackknife**.

hascons, tsscons, vce(), beta, noheader, notable, plus, depname(), mse1, and weights are not allowed with the svy prefix; see [SVY] **svy**.

aweights, fweights, iweights, and pweights are allowed; see **[U] 11.1.6 weight**.

fweights, pweights, and iweights are not allowed with vce(hac *hacspec*).

pweights are not allowed with vce(hc3).

noheader, notable, plus, mse1, and coeflegend do not appear in the dialog box.

See **[U] 20 Estimation and postestimation commands** for more capabilities of estimation commands.

Options

Model

noconstant; see [R] **Estimation options**.

hascons indicates that a user-defined constant or its equivalent is specified among the independent variables in *indepvars*. Some caution is recommended when specifying this option, as resulting estimates may not be as accurate as they otherwise would be. Use of this option requires "sweeping" the constant last, so the moment matrix must be accumulated in absolute rather than deviation form. This option may be safely specified when the means of the dependent and independent variables are all reasonable and there is not much collinearity between the independent variables. The best procedure is to view hascons as a reporting option—estimate with and without hascons and verify that the coefficients and standard errors of the variables not affected by the identity of the constant are unchanged.

tsscons forces the total sum of squares to be computed as though the model has a constant, that is, as deviations from the mean of the dependent variable. This is a rarely used option that has an effect only when specified with noconstant. It affects the total sum of squares and all results derived from the total sum of squares.

SE/Robust

vce(*vcetype*) specifies the type of standard error reported, which includes types that are derived from asymptotic theory (ols), that are robust to some kinds of misspecification (robust), that allow for intragroup correlation (cluster *clustvarlist*), and that use bootstrap or jackknife methods (bootstrap, jackknife); see [R] ***vce_option***.

vce(ols), the default, uses the standard homoskedastic variance estimator for ordinary least-squares regression.

vce(robust) specifies that standard errors allow for heteroskedasticity in the variance of the observations. vce(robust) uses $\hat{\sigma}_j^2 = \{N/(N-k)\}u_j^2$ as an estimate of the variance of the jth observation, where N is the number of observations, k is the number of regressors, u_j is the calculated residual, and $N/(N-k)$ is included to improve the overall estimate's small-sample properties.

vce(cluster *clustvarlist)* specifies that standard errors allow for intragroup correlation within groups defined by one or more variables in *clustvarlist*, relaxing the usual requirement that the observations be independent. For example, vce(cluster clustvar1) produces cluster–robust standard errors that allow for observations that are independent across groups defined by clustvar1 but not necessarily independent within groups. You could also type vce(cluster clustvar1 clustvar2 ... cluster$p$) to account for correlation within groups formed by p variables (multiway clustering).

regress also allows the following:

vce(hc2 [*clustvar*] [, dfadjust]) and vce(hc3 [*clustvar*] [, dfadjust hansen]) specify alternative bias corrections for the heteroskedasticity-robust and cluster–robust variance calculations.

vce(hc2) uses $u_j^2/(1-h_{jj})$ as the observation's variance estimate, where h_{jj} is the diagonal element of the hat (projection) matrix. This estimate is unbiased if the model really is homoskedastic. vce(hc2) tends to produce slightly more conservative confidence intervals than vce(robust). vce(hc2 *clustvar*) produces estimates that allow for intragroup correlation within groups defined by *clustvar*. dfadjust computes Bell and McCaffrey's (2002) adjusted degrees of freedom based on *clustvar*.

vce(hc3) uses $u_j^2/(1-h_{jj})^2$ as the observation's variance estimate. Davidson and MacKinnon (1993) report that this bias-correction method tends to produce better results when the model really is heteroskedastic. vce(hc3) produces confidence intervals that tend to be even more conservative than vce(hc2) and vce(robust).

vce(hc3 *clustvar*) is part of StataNow. It produces estimates that allow for intragroup correlation within groups defined by *clustvar*. dfadjust computes Bell and McCaffrey's (2002) adjusted degrees of freedom based on *clustvar*. hansen also computes the adjusted degrees of freedom and additionally applies a scaling factor when computing p-values and confidence intervals as described in Hansen (2025b). hansen implies dfadjust.

See Davidson and MacKinnon (1993, 554–556) and Angrist and Pischke (2009, 294–308) for more discussion on these two bias corrections.

Note that dfadjust and hansen do not affect multiple-imputation results when the command is used with mi estimate. See *Methods and formulas* for a description of the computation of these adjustments.

vce(hac *hacspec*) is part of StataNow. It requests a heteroskedasticity- and autocorrelation-consistent (HAC) variance–covariance matrix. The full syntax of *hacspec* is one of the following:

vce(hac *kernel* [#]) requests a HAC variance–covariance matrix using the specified kernel (see below) with optional # lags. The bandwidth of a kernel is equal to # + 1. If # is not specified, a kernel with $N-2$ lags is used, where N is the sample size.

vce(hac *kernel* opt [#]) requests a HAC variance–covariance matrix using the specified kernel (see below), and the lag order is selected using Newey and West's (1994) optimal lag-selection algorithm. # is an optional tuning parameter that affects the lag order selected; see *Methods and formulas*.

kernel may be one of the following:

bartlett or nwest requests the Bartlett (Newey–West) kernel.

parzen or gallant requests the Parzen (Gallant 1987) kernel.

quadraticspectral or andrews requests the quadratic spectral (Andrews 1991) kernel.

You must tsset your data before specifying regress with vce(hac *hacspec*).

Reporting

level(#); see [R] **Estimation options**.

beta asks that standardized beta coefficients be reported instead of confidence intervals. The beta coefficients are the regression coefficients obtained by first standardizing all variables to have a mean of 0 and a standard deviation of 1. beta may not be specified with vce(cluster *clustvarlist*) or the svy prefix.

eform(*string*) is used only in programs and ado-files that use regress to fit models other than linear regression. eform() specifies that the coefficient table be displayed in exponentiated form as defined in [R] **Maximize** and that *string* be used to label the exponentiated coefficients in the table.

depname(*varname*) is used only in programs and ado-files that use regress to fit models other than linear regression. depname() may be specified only at estimation time. *varname* is recorded as the identity of the dependent variable, even though the estimates are calculated using *depvar*. This method affects the labeling of the output—not the results calculated—but could affect subsequent

calculations made by predict, where the residual would be calculated as deviations from *varname* rather than *depvar*. depname() is most typically used when *depvar* is a temporary variable (see [P] **macro**) used as a proxy for *varname*.

depname() is not allowed with the svy prefix.

clustertable displays a table reporting cluster combinations and the number of clusters per combination. This option is available only when vce(cluster *clustvarlist*) is specified with more than one variable in *clustvarlist* to compute multiway cluster–robust standard errors.

display_options: noci, nopvalues, dfci, dfpvalues, noomitted, vsquish, noemptycells, baselevels, allbaselevels, nofvlabel, fvwrap(#), fvwrapon(*style*), cformat(%*fmt*), pformat(%*fmt*), sformat(%*fmt*), and nolstretch; see [R] **Estimation options**.

dfci specifies that parameter degrees of freedom and confidence intervals be reported in the coefficient table.

dfpvalues specifies that parameter degrees of freedom and p-values be reported in the coefficient table.

The following options are available with regress but are not shown in the dialog box:

noheader suppresses the display of the ANOVA table and summary statistics at the top of the output; only the coefficient table is displayed. This option is often used in programs and ado-files.

notable suppresses display of the coefficient table.

plus specifies that the output table be made extendable. This option is often used in programs and ado-files.

mse1 is used only in programs and ado-files that use regress to fit models other than linear regression and is not allowed with the svy prefix. mse1 sets the mean squared error to 1, forcing the variance–covariance matrix of the estimators to be $(\mathbf{X}'\mathbf{X})^{-1}$ (see *Methods and formulas* below) and affecting calculated standard errors. Degrees of freedom for t statistics is calculated as N rather than $N - k$.

coeflegend; see [R] **Estimation options**.

Remarks and examples

Remarks are presented under the following headings:

Ordinary least squares
Treatment of the constant
Robust standard errors
Weighted regression
Video examples

regress performs linear regression, including ordinary least squares and weighted least squares. See **[U] 27 Overview of Stata estimation commands** for a list of other regression commands that may be of interest. For a general discussion of linear regression, see Kutner et al. (2005).

See Stock and Watson (2019) and Wooldridge (2020) for an excellent treatment of estimation, inference, interpretation, and specification testing in linear regression models. See Wooldridge (2010, chap. 4) for a more advanced discussion along the same lines.

See Hamilton (2013, chap. 7) and Cameron and Trivedi (2022, chap. 3) for an introduction to linear regression using Stata. Dohoo, Martin, and Stryhn (2012, 2010) discuss linear regression using examples from epidemiology, and Stata datasets and do-files used in the text are available. Cameron and Trivedi (2022) discuss linear regression using econometric examples with Stata. Mitchell (2021) shows how to use graphics and postestimation commands to understand a fitted regression model.

Chatterjee and Hadi (2012) explain regression analysis by using examples containing typical problems that you might encounter when performing exploratory data analysis. We also recommend Weisberg (2014), who emphasizes the importance of the assumptions of linear regression and problems resulting from these assumptions. Becketti (2020) discusses regression analysis with an emphasis on time-series data. Angrist and Pischke (2009) approach regression as a tool for exploring relationships, estimating treatment effects, and providing answers to public policy questions. For a mathematically rigorous treatment, see Peracchi (2001, chap. 6). Finally, see Plackett (1972) if you are interested in the history of regression. Least squares, which dates back to the 1790s, was discovered independently by Legendre and Gauss.

Ordinary least squares

▷ Example 1: Basic linear regression

Suppose that we have data on the mileage rating and weight of 74 automobiles. The variables in our data are `mpg`, `weight`, and `foreign`. The last variable assumes the value 1 for foreign and 0 for domestic automobiles. We wish to fit the model

$$\texttt{mpg} = \beta_0 + \beta_1\texttt{weight} + \beta_2\texttt{foreign} + \epsilon$$

This model can be fit with `regress` by typing

```
. use https://www.stata-press.com/data/r19/auto
(1978 automobile data)
. regress mpg weight foreign
```

Source	SS	df	MS	Number of obs	=	74
				F(2, 71)	=	69.75
Model	1619.2877	2	809.643849	Prob > F	=	0.0000
Residual	824.171761	71	11.608053	R-squared	=	0.6627
				Adj R-squared	=	0.6532
Total	2443.45946	73	33.4720474	Root MSE	=	3.4071

mpg	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
weight	-.0065879	.0006371	-10.34	0.000	-.0078583	-.0053175
foreign	-1.650029	1.075994	-1.53	0.130	-3.7955	.4954422
_cons	41.6797	2.165547	19.25	0.000	37.36172	45.99768

`regress` produces a variety of summary statistics along with the table of regression coefficients. At the upper left, `regress` reports an analysis-of-variance (ANOVA) table. The column headings `SS`, `df`, and `MS` stand for "sum of squares", "degrees of freedom", and "mean square", respectively. In this example, the total sum of squares is 2,443.5: 1,619.3 accounted for by the model and 824.2 left unexplained. Because the regression included a constant, the total sum reflects the sum after removal of means, as does the sum of squares due to the model. The table also reveals that there are 73 total degrees of freedom (counted as 74 observations less 1 for the mean removal), of which 2 are consumed by the model, leaving 71 for the residual.

To the right of the ANOVA table are presented other summary statistics. The F statistic associated with the ANOVA table is 69.75. The statistic has 2 numerator and 71 denominator degrees of freedom. The F statistic tests the hypothesis that all coefficients excluding the constant are zero. The chance of observing an F statistic that large or larger is reported as 0.0000, which is Stata's way of indicating a number smaller than 0.00005. The R^2 for the regression is 0.6627, and the R^2 adjusted for degrees of freedom (R_a^2) is 0.6532. The root mean squared error, labeled `Root MSE`, is 3.4071. It is the square root of the mean squared error reported for the residual in the ANOVA table.

Finally, Stata produces a table of the estimated coefficients. The first line of the table indicates that the left-hand-side variable is `mpg`. Thereafter follow the estimated coefficients. Our fitted model is

$$\texttt{mpg_hat} = 41.68 - 0.0066\,\texttt{weight} - 1.65\,\texttt{foreign}$$

Reported to the right of the coefficients in the output are the standard errors. For instance, the standard error for the coefficient on `weight` is 0.0006371. The corresponding t statistic is -10.34, which has a two-sided significance level of 0.000. This number indicates that the significance is less than 0.0005. The 95% confidence interval for the coefficient is $[\,-0.0079, -0.0053\,]$.

◁

▷ Example 2: Transforming the dependent variable

If we had a graph comparing `mpg` with `weight`, we would notice that the relationship is distinctly nonlinear. This is to be expected because energy usage per distance should increase linearly with weight, but `mpg` is measuring distance per energy used. We could obtain a better model by generating a new variable measuring the number of gallons used per 100 miles (`gp100m`) and then using this new variable in our model:

$$\texttt{gp100m} = \beta_0 + \beta_1\texttt{weight} + \beta_2\texttt{foreign} + \epsilon$$

We can now fit this model:

```
. generate gp100m = 100/mpg
. regress gp100m weight foreign
```

Source	SS	df	MS	Number of obs	=	74
				F(2, 71)	=	113.97
Model	91.1761694	2	45.5880847	Prob > F	=	0.0000
Residual	28.4000913	71	.400001287	R-squared	=	0.7625
				Adj R-squared	=	0.7558
Total	119.576261	73	1.63803097	Root MSE	=	.63246

gp100m	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
weight	.0016254	.0001183	13.74	0.000	.0013896	.0018612
foreign	.6220535	.1997381	3.11	0.003	.2237871	1.02032
_cons	-.0734839	.4019932	-0.18	0.855	-.8750354	.7280677

Fitting the physically reasonable model increases our R^2 to 0.7625.

◁

▷ Example 3: Obtaining beta coefficients

`regress` shares the features of all estimation commands. Among other things, this means that after running a regression, we can use `test` to test hypotheses about the coefficients, `estat vce` to examine the covariance matrix of the estimators, and `predict` to obtain predicted values, residuals, and influence statistics. See **[U] 20 Estimation and postestimation commands**. Options that affect how estimates are displayed, such as `beta` or `level()`, can be used when replaying results.

Suppose that we meant to specify the `beta` option to obtain beta coefficients (regression coefficients normalized by the ratio of the standard deviation of the regressor to the standard deviation of the dependent variable). Even though we forgot, we can specify the option now:

```
. regress, beta
```

Source	SS	df	MS	Number of obs	=	74
				F(2, 71)	=	113.97
Model	91.1761694	2	45.5880847	Prob > F	=	0.0000
Residual	28.4000913	71	.400001287	R-squared	=	0.7625
				Adj R-squared	=	0.7558
Total	119.576261	73	1.63803097	Root MSE	=	.63246

gp100m	Coefficient	Std. err.	t	P>|t|	Beta
weight	.0016254	.0001183	13.74	0.000	.9870255
foreign	.6220535	.1997381	3.11	0.003	.2236673
_cons	-.0734839	.4019932	-0.18	0.855	.

◁

Treatment of the constant

By default, `regress` includes an intercept (constant) term in the model. The `noconstant` option suppresses it, and the `hascons` option tells `regress` that the model already has one.

▷ Example 4: Suppressing the constant term

We wish to fit a regression of the `weight` of an automobile against its `length`, and we wish to impose the constraint that the weight is zero when the length is zero.

If we simply type `regress weight length`, we are fitting the model

$$\texttt{weight} = \beta_0 + \beta_1\,\texttt{length} + \epsilon$$

Here a `length` of zero corresponds to a `weight` of β_0. We want to force β_0 to be zero or, equivalently, estimate an equation that does not include an intercept:

$$\texttt{weight} = \beta_1\,\texttt{length} + \epsilon$$

We do this by specifying the noconstant option:

```
. regress weight length, noconstant
```

Source	SS	df	MS	Number of obs	=	74
				F(1, 73)	=	3450.13
Model	703869302	1	703869302	Prob > F	=	0.0000
Residual	14892897.8	73	204012.299	R-squared	=	0.9793
				Adj R-squared	=	0.9790
Total	718762200	74	9713002.7	Root MSE	=	451.68

weight	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
length	16.29829	.2774752	58.74	0.000	15.74528	16.8513

In our data, length is measured in inches and weight in pounds. We discover that each inch of length adds 16 pounds to the weight.

◁

Sometimes there is no need for Stata to include a constant term in the model. Most commonly, this occurs when the model contains a set of mutually exclusive indicator variables. hascons is a variation of the noconstant option—it tells Stata not to add a constant to the regression because the regression specification already has one, either directly or indirectly.

For instance, we now refit our model of weight as a function of length and include separate constants for foreign and domestic cars by specifying bn.foreign. bn.foreign is factor-variable notation for "no base for foreign" or "include all levels of variable foreign in the model"; see **[U] 11.4.3 Factor variables**.

```
. regress weight length bn.foreign, hascons
```

Source	SS	df	MS	Number of obs	=	74
				F(2, 71)	=	316.54
Model	39647744.7	2	19823872.3	Prob > F	=	0.0000
Residual	4446433.7	71	62625.8268	R-squared	=	0.8992
				Adj R-squared	=	0.8963
Total	44094178.4	73	604029.841	Root MSE	=	250.25

weight	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
length	31.44455	1.601234	19.64	0.000	28.25178	34.63732
foreign						
Domestic	-2850.25	315.9691	-9.02	0.000	-3480.274	-2220.225
Foreign	-2983.927	275.1041	-10.85	0.000	-3532.469	-2435.385

❑ Technical note

There is a subtle distinction between the hascons and noconstant options. We can most easily reveal it by refitting the last regression, specifying noconstant rather than hascons:

```
. regress weight length bn.foreign, noconstant
```

Source	SS	df	MS	Number of obs	=	74
				F(3, 71)	=	3802.03
Model	714315766	3	238105255	Prob > F	=	0.0000
Residual	4446433.7	71	62625.8268	R-squared	=	0.9938
				Adj R-squared	=	0.9936
Total	718762200	74	9713002.7	Root MSE	=	250.25

weight	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
length	31.44455	1.601234	19.64	0.000	28.25178	34.63732
foreign						
Domestic	-2850.25	315.9691	-9.02	0.000	-3480.274	-2220.225
Foreign	-2983.927	275.1041	-10.85	0.000	-3532.469	-2435.385

Comparing this output with that produced by the previous regress command, we see that they are almost, but not quite, identical. The parameter estimates and their associated statistics—the second half of the output—are identical. The overall summary statistics and the ANOVA table—the first half of the output—are different, however.

In the first case, the R^2 is shown as 0.8992; here it is shown as 0.9938. In the first case, the F statistic is 316.54; now it is 3,802.03. The numerator degrees of freedom is different as well. In the first case, the numerator degrees of freedom is 2; now the degrees of freedom is 3. Which is correct?

Both are. Specifying the hascons option causes regress to adjust the ANOVA table and its associated statistics for the explanatory power of the constant. The regression in effect has a constant; it is just written in such a way that a separate constant is unnecessary. No such adjustment is made with the noconstant option.

❑

❑ Technical note

When the hascons option is specified, regress checks to make sure that the model does in fact have a constant term. If regress cannot find a constant term, it automatically adds one. Fitting a model of weight on length and specifying the hascons option, we obtain

```
. regress weight length, hascons
note: option hascons false.
```

Source	SS	df	MS	Number of obs	=	74
				F(1, 72)	=	613.27
Model	39461306.8	1	39461306.8	Prob > F	=	0.0000
Residual	4632871.55	72	64345.4382	R-squared	=	0.8949
				Adj R-squared	=	0.8935
Total	44094178.4	73	604029.841	Root MSE	=	253.66

weight	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
length	33.01988	1.333364	24.76	0.000	30.36187	35.67789
_cons	-3186.047	252.3113	-12.63	0.000	-3689.02	-2683.073

Even though we specified hascons, regress included a constant, anyway. It also added a note to our output: "note: option hascons false".

❑

❑ Technical note

Even if the model specification effectively includes a constant term, we need not specify the hascons option. regress is always on the lookout for collinear variables and omits them from the model. For instance,

```
. regress weight length bn.foreign
note: 1.foreign omitted because of collinearity.
```

Source	SS	df	MS	Number of obs	=	74
				F(2, 71)	=	316.54
Model	39647744.7	2	19823872.3	Prob > F	=	0.0000
Residual	4446433.7	71	62625.8268	R-squared	=	0.8992
				Adj R-squared	=	0.8963
Total	44094178.4	73	604029.841	Root MSE	=	250.25

weight	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
length	31.44455	1.601234	19.64	0.000	28.25178	34.63732
foreign						
Domestic	133.6775	77.47615	1.73	0.089	-20.80555	288.1605
Foreign	0	(omitted)				
_cons	-2983.927	275.1041	-10.85	0.000	-3532.469	-2435.385

❑

Robust standard errors

regress with the vce(robust) option substitutes a robust variance matrix calculation for the conventional calculation, or if vce(cluster *clustvarlist*) is specified, allows relaxing the assumption of independence within groups. How this method works is explained in [U] **20.22 Obtaining robust variance estimates**. Below, we show how well this approach works.

▷ Example 5: Heteroskedasticity and robust standard errors

Specifying the vce(robust) option is equivalent to requesting White-corrected standard errors in the presence of heteroskedasticity. We use the automobile data and, in the process of looking at the energy efficiency of cars, analyze a variable with considerable heteroskedasticity.

We will examine the amount of energy—measured in gallons of gasoline—that the cars in the data need to move 1,000 pounds of their weight 100 miles. We are going to examine the relative efficiency of foreign and domestic cars.

```
. generate gpmw = ((1/mpg)/weight)*100*1000
. summarize gpmw
```

Variable	Obs	Mean	Std. dev.	Min	Max
gpmw	74	1.682184	.2426311	1.09553	2.30521

In these data, the engines consume between 1.10 and 2.31 gallons of gas to move 1,000 pounds of the car's weight 100 miles. If we ran a regression with conventional standard errors of gpmw on foreign, we would obtain

```
. regress gpmw foreign
```

Source	SS	df	MS	Number of obs	=	74
				F(1, 72)	=	20.07
Model	.936705572	1	.936705572	Prob > F	=	0.0000
Residual	3.36079459	72	.046677703	R-squared	=	0.2180
				Adj R-squared	=	0.2071
Total	4.29750017	73	.058869865	Root MSE	=	.21605

gpmw	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
foreign	.2461526	.0549487	4.48	0.000	.1366143	.3556909
_cons	1.609004	.0299608	53.70	0.000	1.549278	1.66873

regress with the vce(robust) option, on the other hand, reports

```
. regress gpmw foreign, vce(robust)
```

Linear regression

				Number of obs	=	74
				F(1, 72)	=	13.13
				Prob > F	=	0.0005
				R-squared	=	0.2180
				Root MSE	=	.21605

gpmw	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
foreign	.2461526	.0679238	3.62	0.001	.1107489	.3815563
_cons	1.609004	.0234535	68.60	0.000	1.56225	1.655758

The point estimates are the same (foreign cars need one-quarter gallon more gas), but the standard errors differ by roughly 20%. Conventional regression reports the 95% confidence interval as $[0.14, 0.36]$, whereas the robust standard errors make the interval $[0.11, 0.38]$.

Which is right? Notice that gpmw is a variable with considerable heteroskedasticity:

```
. tabulate foreign, summarize(gpmw)
```

Car origin	Summary of gpmw Mean	Std. dev.	Freq.
Domestic	1.6090039	.16845182	52
Foreign	1.8551565	.30186861	22
Total	1.6821844	.24263113	74

Thus, here we favor the robust standard errors. In **[U] 20.22 Obtaining robust variance estimates**, we show another example using linear regression where it makes little difference whether we specify vce(robust). The linear-regression assumptions were true, and we obtained nearly linear-regression results. The advantage of the robust estimate is that in neither case did we have to check assumptions.

◁

❑ Technical note

`regress` purposefully suppresses displaying the ANOVA table when `vce(robust)` is specified. This is done because the sums of squares are no longer appropriate for use in the usual hypothesis tests, even though computationally the sums of squares remain the same. In the nonrobust setting, the F statistic reported by `regress` is defined in terms of the sums of squares, as in ANOVA. When `vce(robust)` is specified, the ANOVA test is not valid, and the F statistic corresponds to a Wald test based on the robustly estimated variance matrix.

Some references give formulas for the F statistic in terms of either R^2 or the root MSE. It is not appropriate to use those formulas for the F statistic with robust standard errors because the R^2 and root MSE are calculated from the sums of squares. Moreover, the root MSE can no longer be used as an estimate for σ because there is no longer a single σ to estimate—the variance of the residual varies observation by observation. However, `regress` continues to report the R^2 and the root MSE in the robust setting because those statistics are still usable in other settings. In particular, R^2 remains valid as a goodness-of-fit statistic.

❑

▷ Example 6: Alternative robust standard errors

The `vce(hc2)` and `vce(hc3)` options modify the robust variance calculation. In the context of linear regression without clustering, the idea behind the robust calculation is somehow to measure σ_j^2, the variance of the residual associated with the jth observation, and then to use that estimate to improve the estimated variance of $\widehat{\boldsymbol{\beta}}$. Because residuals have (theoretically and practically) mean 0, one estimate of σ_j^2 is the observation's squared residual itself—u_j^2. A finite-sample correction could improve that by multiplying u_j^2 by $n/(n-k)$, and, as a matter of fact, `vce(robust)` uses $\{n/(n-k)\}u_j^2$ as its estimate of the residual's variance.

`vce(hc2)` and `vce(hc3)` use alternative estimators of the observation-specific variances. For instance, if the residuals are homoskedastic, we can show that the expected value of u_j^2 is $\sigma^2(1-h_{jj})$, where h_{jj} is the jth diagonal element of the projection (hat) matrix. h_{jj} has average value k/n, so $1-h_{jj}$ has average value $1-k/n = (n-k)/n$. Thus, the default robust estimator $\widehat{\sigma}_j = \{n/(n-k)\}u_j^2$ amounts to dividing u_j^2 by the average of the expectation.

`vce(hc2)` divides u_j^2 by $1-h_{jj}$ itself, so it should yield better estimates if the residuals really are homoskedastic. `vce(hc3)` divides u_j^2 by $(1-h_{jj})^2$ and has no such clean interpretation. Davidson and MacKinnon (1993) show that $u_j^2/(1-h_{jj})^2$ approximates a more complicated estimator that they obtain by jackknifing (MacKinnon and White 1985). Angrist and Pischke (2009) also illustrate the relative merits of these adjustments.

Here are the results of refitting our efficiency model using `vce(hc2)` and `vce(hc3)`:

```
. regress gpmw foreign, vce(hc2)
Linear regression                               Number of obs     =         74
                                                F(1, 72)          =      12.93
                                                Prob > F          =     0.0006
                                                R-squared         =     0.2180
                                                Root MSE          =     .21605
------------------------------------------------------------------------------
             |               Robust HC2
        gpmw | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     foreign |   .2461526   .0684669     3.60   0.001     .1096662    .3826389
       _cons |   1.609004   .0233601    68.88   0.000     1.562437    1.655571
------------------------------------------------------------------------------

. regress gpmw foreign, vce(hc3)
Linear regression                               Number of obs     =         74
                                                F(1, 72)          =      12.38
                                                Prob > F          =     0.0008
                                                R-squared         =     0.2180
                                                Root MSE          =     .21605
------------------------------------------------------------------------------
             |               Robust HC3
        gpmw | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     foreign |   .2461526    .069969     3.52   0.001     .1066719    .3856332
       _cons |   1.609004   .023588    68.21   0.000     1.561982    1.656026
------------------------------------------------------------------------------
```

◁

▷ Example 7: Standard errors for clustered data

The `vce(cluster` *clustvarlist*`)`, `vce(hc2` *clustvar*`)`, and `vce(hc3` *clustvar*`)` options relax the assumption of independence. Below, we have 28,534 observations on 4,711 women aged 14–46 years. Data were collected on these women between 1968 and 1988. We are going to fit a classic earnings model, and we begin by ignoring that the majority of the women in the dataset have multiple observations.

```
. use https://www.stata-press.com/data/r19/regsmpl, clear
(NLS women 14-26 in 1968)
. regress ln_wage age c.age#c.age tenure
      Source |       SS           df       MS      Number of obs   =    28,101
-------------+----------------------------------   F(3, 28097)     =   1842.45
       Model |  1054.52501         3  351.508335   Prob > F        =    0.0000
    Residual |  5360.43962    28,097  .190783344   R-squared       =    0.1644
-------------+----------------------------------   Adj R-squared   =    0.1643
       Total |  6414.96462    28,100  .228290556   Root MSE        =    .43679
------------------------------------------------------------------------------
     ln_wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0752172   .0034736    21.65   0.000     .0684088    .0820257
             |
 c.age#c.age |  -.0010851   .0000575   -18.86   0.000    -.0011979   -.0009724
             |
      tenure |   .0390877   .0007743    50.48   0.000     .0375699    .0406054
       _cons |   .3339821   .0504413     6.62   0.000     .2351148    .4328495
------------------------------------------------------------------------------
```

The number of observations in our model is 28,101 because Stata drops observations that have a missing value for one or more of the variables in the model. We can be reasonably certain that the standard errors reported above are meaningless. Without a doubt, a woman with higher-than-average wages in one year typically has higher-than-average wages in other years, and so the residuals are not independent. One way to deal with this is to use cluster–robust standard errors. We do this by specifying `vce(cluster id)`, `vce(hc2 id)`, or `vce(hc3 id)`, which treat only observations with different person `id`s as truly independent:

```
. regress ln_wage age c.age#c.age tenure, vce(cluster id)
Linear regression                               Number of obs     =     28,101
                                                F(3, 4698)        =     748.82
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1644
                                                Root MSE          =     .43679

                                  (Std. err. adjusted for 4,699 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0752172   .0045711    16.45   0.000     .0662557    .0841788
             |
 c.age#c.age |  -.0010851   .0000778   -13.94   0.000    -.0012377   -.0009325
             |
      tenure |   .0390877   .0014425    27.10   0.000     .0362596    .0419157
       _cons |   .3339821   .0641918     5.20   0.000      .208136    .4598282
------------------------------------------------------------------------------
```

For comparison, we focus on the tenure coefficient, which in economics jargon can be interpreted as the rate of return for keeping your job. The 95% confidence interval we previously estimated—an interval we do not believe—is $[0.038, 0.041]$. The robust interval is twice as wide, being $[0.036, 0.042]$. For this example, `vce(hc2 id)` and `vce(hc3 id)` give standard errors similar to `vce(cluster id)`.

Another possible way to account for the lack of independence is to fit a random-effects model. Here is the random-effects result:

```
. xtreg ln_wage age c.age#c.age tenure, re
Random-effects GLS regression                   Number of obs     =     28,101
Group variable: idcode                          Number of groups  =      4,699

R-squared:                                      Obs per group:
     Within  = 0.1370                                         min =          1
     Between = 0.2154                                         avg =        6.0
     Overall = 0.1608                                         max =         15

                                                Wald chi2(3)      =    4717.05
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =     0.0000

------------------------------------------------------------------------------
     ln_wage |  Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0568296   .0026958    21.08   0.000     .0515459    .0621132
             |
 c.age#c.age |  -.0007566   .0000447   -16.93   0.000    -.0008441    -.000669
             |
      tenure |   .0260135   .0007477    34.79   0.000     .0245481    .0274789
       _cons |   .6136792   .0394611    15.55   0.000     .5363368    .6910216
-------------+----------------------------------------------------------------
     sigma_u |  .33542449
     sigma_e |  .29674679
         rho |  .56095413   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

Robust regression estimated the 95% interval [0.036, 0.042], and `xtreg` (see [XT] **xtreg**) estimates [0.025, 0.027]. Which is better? The random-effects regression estimator assumes a lot. We can check some of these assumptions by performing a Hausman test. Using `estimates` (see [R] **estimates store**), we store the random-effects estimation results, and then we run the required fixed-effects regression to perform the test.

```
. estimates store random
. xtreg ln_wage age c.age#c.age tenure, fe
Fixed-effects (within) regression               Number of obs      =     28,101
Group variable: idcode                          Number of groups   =      4,699
R-squared:                                      Obs per group:
     Within  = 0.1375                                         min =          1
     Between = 0.2066                                         avg =        6.0
     Overall = 0.1568                                         max =         15
                                                F(3, 23399)        =    1243.00
corr(u_i, Xb) = 0.1380                          Prob > F           =     0.0000
```

ln_wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
age	.0522751	.002783	18.78	0.000	.0468202	.05773
c.age#c.age	-.0006717	.0000461	-14.56	0.000	-.0007621	-.0005813
tenure	.021738	.000799	27.21	0.000	.020172	.023304
_cons	.687178	.0405944	16.93	0.000	.6076103	.7667456
sigma_u	.38743138					
sigma_e	.29674679					
rho	.6302569	(fraction of variance due to u_i)				

```
F test that all u_i=0: F(4698, 23399) = 7.98                 Prob > F = 0.0000
. hausman . random
```

	Coefficients (b)	Coefficients (B)	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
	.	random		
age	.0522751	.0568296	-.0045545	.0006913
c.age#c.age	-.0006717	-.0007566	.0000849	.0000115
tenure	.021738	.0260135	-.0042756	.0002816

```
                b = Consistent under H0 and Ha; obtained from xtreg.
     B = Inconsistent under Ha, efficient under H0; obtained from xtreg.
Test of H0: Difference in coefficients not systematic
    chi2(3) = (b-B)'[(V_b-V_B)^(-1)](b-B)
            = 336.62
Prob > chi2 = 0.0000
```

The Hausman test casts grave suspicions on the random-effects model we just fit, so we should be careful in interpreting those results.

Meanwhile, our robust regression results still stand, as long as we are careful about the interpretation. The correct interpretation is that, if the data collection were repeated (on women sampled the same way as in the original sample), and if we were to refit the model, 95% of the time we would expect the estimated coefficient on tenure to be in the range [0.036, 0.042].

Even with robust regression, we must be careful about going beyond that statement. Here the Hausman test is probably picking up something that differs within and between person, which would cast doubt on our robust regression model in terms of interpreting $[\,0.036, 0.042\,]$ to contain the rate of return for keeping a job, economywide, for all women, without exception.

Let's take this example a bit further by also recognizing workers with the same education level, `grade`, may be more alike than those with different education levels. Here we will use multiway clustering (Cameron, Gelbach, and Miller 2008) to account for correlations within individuals over years and within education levels, assuming observations from different people and different education level are independent.

```
. regress ln_wage age c.age#c.age tenure, vce(cluster idcode grade) clustertable
Linear regression                               Number of obs = 28,099
Clusters per comb.:                             Cluster comb. =      3
  min =    19                                   F(3, 18)      = 247.23
  avg = 3,138                                   Prob > F      = 0.0000
  max = 4,697                                   R-squared     = 0.1644
                                                Adj R-squared = 0.1644
                                                Root MSE      = 0.4368

                    Clusters
Cluster combination per comb.
-------------------------------
             idcode     4,697
              grade        19
      idcode#grade     4,697
-------------------------------

                                (Std. err. adjusted for multiway clustering)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0751665   .018361     4.09   0.001     .0365914    .1137415
             |
 c.age#c.age |  -.0010842   .0002504   -4.33   0.000    -.0016103   -.0005581
             |
      tenure |   .0391104   .0018302   21.37   0.000     .0352653    .0429554
       _cons |    .334631   .2769961    1.21   0.243    -.2473162    .9165782
------------------------------------------------------------------------------
Cluster combinations formed by idcode and grade.
```

With the `clustertable` option, the output includes a table that describes the cluster combinations and reports the number of levels for each cluster combination. We clustered on two variables, so there are three cluster combinations, and, in general, for p cluster variables, there are $p^2 - 1$ cluster combinations. The t-statistic degrees of freedom is chosen from the cluster combination that has the smallest number of levels. In this case, we have $19 - 1 = 18$ degrees of freedom, and the Wald test F statistic, computed from the cluster–robust VCE, has a denominator degrees of freedom of 18. The 95% confidence interval for `tenure` increases slightly to $[\,0.035, 0.043\,]$. Yet the confidence intervals here are not strictly comparable; this model is fit to two fewer observations because of missing values in the cluster variable `grade`, and we conjecture that observations are correlated within `grade`.

◁

Weighted regression

regress can perform weighted and unweighted regression. We indicate the weight by specifying the [*weight*] qualifier.

▷ Example 8: Using means as regression variables

We have census data recording the deathrate (drate) and median age (medage) for each state. The data also record the region of the country in which each state is located and the overall population of the state:

```
. use https://www.stata-press.com/data/r19/census9
(1980 Census data by state)
. describe
Contains data from https://www.stata-press.com/data/r19/census9.dta
 Observations:            50                  1980 Census data by state
    Variables:             6                  2 Dec 2024 15:22
```

Variable name	Storage type	Display format	Value label	Variable label
state	str13	%-13s		State
state2	str2	%-2s		Two-letter state abbreviation
drate	int	%9.0g		Deathrate
pop	long	%12.0gc		Population
medage	float	%9.2f		Median age
region	byte	%-8.0g	cenreg	Census region

```
Sorted by:
```

We can use factor variables to include dummy variables for region. Because the variables in the regression reflect means rather than individual observations, the appropriate method of estimation is analytically weighted least squares (Davidson and MacKinnon 2004, 261–262), where the weight is total population:

```
. regress drate medage i.region [aweight=pop]
(sum of wgt is 225,907,472)
```

Source	SS	df	MS		
Model	4096.6093	4	1024.15232	Number of obs	= 50
Residual	1238.40987	45	27.5202192	F(4, 45)	= 37.21
				Prob > F	= 0.0000
				R-squared	= 0.7679
				Adj R-squared	= 0.7472
Total	5335.01916	49	108.877942	Root MSE	= 5.246

drate	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
medage	4.283183	.5393329	7.94	0.000	3.196911	5.369455
region						
N Cntrl	.3138738	2.456431	0.13	0.899	-4.633632	5.26138
South	-1.438452	2.320244	-0.62	0.538	-6.111663	3.234758
West	-10.90629	2.681349	-4.07	0.000	-16.30681	-5.505777
_cons	-39.14727	17.23613	-2.27	0.028	-73.86262	-4.431915

To weight the regression by population, we added the qualifier [aweight=pop] to the end of the regress command. Stata informed us that the sum of the weight is 2.2591×10^8; there were approximately 226 million people residing in the United States according to our 1980 data.

In the weighted regression, we see that the coefficient on West is statistically significant but that the coefficients on N Cntrl and South are not. We use testparm to test the joint significance of the region variable. Because we fit a weighted regression, testparm uses the appropriately weighted variance–covariance matrix.

```
. testparm i.region
 ( 1)  2.region = 0
 ( 2)  3.region = 0
 ( 3)  4.region = 0
       F(  3,    45) =    9.84
            Prob > F =    0.0000
```

The results indicate that the region variables are jointly significant. Note that we could have performed this same test by typing contrast region. You may prefer to use the contrast command because, in addition to the joint test, you can perform other tests such as comparisons of each region's mean to the grand mean; see [R] **contrast** for more information.

◁

regress also accepts frequency weights (fweights). Frequency weights are appropriate when the data do not reflect cell means but instead represent replicated observations. Specifying aweights or fweights will not change the parameter estimates, but it will change the corresponding significance levels.

For instance, if we specified [fweight=pop] in the weighted regression example above—which would be statistically incorrect—Stata would treat the data as if the data represented 226 million independent observations on death rates and median age. The data most certainly do not represent that—they represent 50 observations on state averages.

With aweights, Stata treats the number of observations on the process as the number of observations in the data. When we specify fweights, Stata treats the number of observations as if it were equal to the sum of the weights; see *Methods and formulas* below.

❑ Technical note

A frequent inquiry sent to StataCorp Technical Services is to describe the effect of specifying [aweight=*exp*] with regress in terms of transformation of the dependent and independent variables. The mechanical answer is that typing

```
. regress y x1 x2 [aweight=n]
```

is equivalent to fitting the model

$$y_j\sqrt{n_j} = \beta_0\sqrt{n_j} + \beta_1 x_{1j}\sqrt{n_j} + \beta_2 x_{2j}\sqrt{n_j} + u_j\sqrt{n_j}$$

This regression will reproduce the coefficients and covariance matrix produced by the aweighted regression. The mean squared errors (estimates of the variance of the residuals) will, however, be different. The transformed regression reports s_t^2, an estimate of $\mathrm{Var}(u_j\sqrt{n_j})$. The aweighted regression reports s_a^2, an estimate of $\mathrm{Var}(u_j\sqrt{n_j}\sqrt{N/\sum_k n_k})$, where N is the number of observations. Thus,

$$s_a^2 = \frac{N}{\sum_k n_k} s_t^2 = \frac{s_t^2}{\overline{n}} \tag{1}$$

The logic for this adjustment is as follows: Consider the model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

Assume that, were this model fit on individuals, $\text{Var}(u) = \sigma_u^2$, a constant. Assume that individual data are not available; what is available are averages $(\overline{y}_j, \overline{x}_{1j}, \overline{x}_{2j})$ for $j = 1, \ldots, N$, and each average is calculated over n_j observations. Then it is still true that

$$\overline{y}_j = \beta_0 + \beta_1 \overline{x}_{1j} + \beta_2 \overline{x}_{2j} + \overline{u}_j$$

where $\overline{u}_j$ is the average of n_j mean 0, variance σ_u^2 deviates and has variance $\sigma_{\overline{u}}^2 = \sigma_u^2/n_j$. Thus, multiplying through by $\sqrt{n_j}$ produces

$$\overline{y}_j\sqrt{n_j} = \beta_0\sqrt{n_j} + \beta_1\overline{x}_{1j}\sqrt{n_j} + \beta_2\overline{x}_{2j}\sqrt{n_j} + \overline{u}_j\sqrt{n_j}$$

and $\text{Var}(\overline{u}_j\sqrt{n_j}) = \sigma_u^2$. The mean squared error, s_t^2, reported by fitting this transformed regression is an estimate of σ_u^2. The coefficients and covariance matrix could also be obtained by `aweighted regress`. The only difference would be in the reported mean squared error, which from (1) is $\sigma_u^2/\overline{n}$. On average, each observation in the data reflects the averages calculated over $\overline{n} = \sum_k n_k/N$ individuals, and thus this reported mean squared error is the average variance of an observation in the dataset. We can retrieve the estimate of σ_u^2 by multiplying the reported mean squared error by $\overline{n}$.

More generally, `aweights` are used to solve general heteroskedasticity problems. In these cases, we have the model

$$y_j = \beta_0 + \beta_1 x_{1j} + \beta_2 x_{2j} + u_j$$

and the variance of u_j is thought to be proportional to a_j. If the variance is proportional to a_j, it is also proportional to αa_j, where α is any positive constant. Not quite arbitrarily, but with no loss of generality, we could choose $\alpha = \sum_k (1/a_k)/N$, the average value of the inverse of a_j. We can then write $\text{Var}(u_j) = k\alpha a_j \sigma^2$, where k is the constant of proportionality that is no longer a function of the scale of the weights.

Dividing this regression through by the $\sqrt{a_j}$,

$$y_j/\sqrt{a_j} = \beta_0/\sqrt{a_j} + \beta_1 x_{1j}/\sqrt{a_j} + \beta_2 x_{2j}/\sqrt{a_j} + u_j/\sqrt{a_j}$$

produces a model with $\text{Var}(u_j/\sqrt{a_j}) = k\alpha\sigma^2$, which is the constant part of $\text{Var}(u_j)$. This variance is a function of α, the average of the reciprocal weights; if the weights are scaled arbitrarily, then so is this variance.

We can also fit this model by typing

```
. regress y x1 x2 [aweight=1/a]
```

This input will produce the same estimates of the coefficients and covariance matrix; the reported mean squared error is, from (1), $\{N/\sum_k (1/a_k)\}k\alpha\sigma^2 = k\sigma^2$. This variance is independent of the scale of a_j.

❑

Video examples

Simple linear regression in Stata

Fitting and interpreting regression models: Linear regression with categorical predictors

Fitting and interpreting regression models: Linear regression with continuous predictors

Fitting and interpreting regression models: Linear regression with continuous and categorical predictors

Stored results

regress stores the following in e():

Scalars

e(N)	number of observations
e(mss)	model sum of squares
e(df_m)	model degrees of freedom
e(rss)	residual sum of squares
e(df_r)	residual degrees of freedom
e(r2)	R^2
e(r2_a)	adjusted R^2
e(F)	F statistic
e(p)	p-value for model test
e(rmse)	root mean squared error
e(ll)	log likelihood under additional assumption of i.i.d. normal errors
e(ll_0)	log likelihood, constant-only model
e(hac_lag)	HAC lag
e(sum_w)	sum of weights
e(N_clust)	number of clusters
e(rank)	rank of e(V)

Macros

e(cmd)	regress
e(cmdline)	command as typed
e(depvar)	name of dependent variable
e(wtype)	weight type
e(wexp)	weight expression
e(model)	ols
e(title)	title in estimation output when vce() is not ols
e(clustvar)	names of cluster variables
e(cluster#)	cluster combination #
e(vce)	*vcetype* specified in vce()
e(vcetype)	title used to label Std. err.
e(hac_kernel)	HAC kernel
e(properties)	b V
e(estat_cmd)	program used to implement estat
e(predict)	program used to implement predict
e(marginsok)	predictions allowed by margins
e(asbalanced)	factor variables fvset as asbalanced
e(asobserved)	factor variables fvset as asobserved

Matrices

e(b)	coefficient vector
e(V)	variance–covariance matrix of the estimators
e(beta)	standardized coefficients
e(V_modelbased)	model-based variance
e(adj_df)	adjusted degrees of freedom when vce(hc2, dfadjust), vce(hc3, dfadjust), or vce(hc3, hansen) is specified
e(adj_scale)	scale adjustment parameter when vce(hc3, hansen) is specified
e(adj_ci)	adjusted confidence intervals when vce(hc3, hansen) is specified
e(adj_p)	adjusted p-values when vce(hc3, hansen) is specified
e(kcluster)	cluster sizes, multiway clustering

Functions

e(sample)	marks estimation sample

In addition to the above, the following is stored in r():

Matrices
r(table) matrix containing the coefficients with their standard errors, test statistics, p-values, and confidence intervals

Note that results stored in r() are updated when the command is replayed and will be replaced when any r-class command is run after the estimation command.

Methods and formulas

Methods and formulas are presented under the following headings:

Ordinary least-squares regression
ANOVA table
Weighted regression
Heteroskedasticity-robust and cluster–robust variance estimation
General notation
Multiway clustering
Degrees of freedom
Adjusted degrees of freedom
HAC standard errors (StataNow)

Ordinary least-squares regression

Variables printed in lowercase and not boldfaced (for example, x) are scalars. Variables printed in lowercase and boldfaced (for example, $\mathbf{x}$) are column vectors. Variables printed in uppercase and boldfaced (for example, $\mathbf{X}$) are matrices.

Let $\mathbf{X}$ denote the $N \times k$ matrix of observations on the right-hand-side variables, where N is the sample size and k is the number of regressors (including the constant). Let $\mathbf{y}$ be the $N \times 1$ vector of observations on the dependent variable. The ordinary least-squares (OLS) coefficients are given by

$$\widehat{\boldsymbol{\beta}}_{\text{ols}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y}) \tag{2}$$

These coefficients minimize the sum of the squared residuals; that is, $\widehat{\boldsymbol{\beta}}_{\text{ols}} = \text{argmin}_{\boldsymbol{\beta}} \sum_{i=1}^{N} (y_i - \mathbf{x}_i\boldsymbol{\beta})^2$.

Although not shown in the notation, unless hascons is specified, $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{y}$ are accumulated in deviation form, and the coefficient for the constant is calculated separately. Working with vectors and matrices in deviation form improves numerical stability. This demeaning procedure is applied when calculating all statistics listed below.

ANOVA table

The total sum of squares (TSS) equals $\mathbf{y}'\mathbf{y}$ if there is no intercept and $\mathbf{y}'\mathbf{y} - \{(\mathbf{1}'\mathbf{y})^2/N\}$ otherwise. The degrees of freedom associated with it is $N - c$, where N is the number of observations and $c = 1$ if there is a constant in the regression and 0 otherwise.

The residual sum of squares (RSS) is defined as $(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}_{\text{ols}})'(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}_{\text{ols}})$. The degrees of freedom associated with it is $N - k$.

The model sum of squares (MSS) equals TSS − RSS. The degrees of freedom associated with it is $k - c$.

The mean squared error, s^2, is defined as $\text{RSS}/(N - k)$. The root mean squared error is s, its square root.

The F statistic with $k-c$ and $N-k$ degrees of freedom is defined as

$$F = \frac{\text{MSS}}{(k-c)s^2}$$

The R^2 is defined as $R^2 = 1 - \text{RSS}/\text{TSS}$.

The adjusted R^2 is defined as $R^2_a = 1 - (1-R^2)(N-c)/(N-k)$.

The conventional homoskedastic estimate of variance is

$$\widehat{\mathbf{V}}_{\text{hom}} = s^2(\mathbf{X}'\mathbf{X})^{-1} \tag{3}$$

The calculation of variance estimates when heteroskedasticity-robust or cluster–robust variance estimates are specified is described below.

Weighted regression

For a given $N \times 1$ vector of weights, $\mathbf{w}$, the weighted least squares (WLS) regression coefficient is defined as

$$\widehat{\boldsymbol{\beta}}_{\text{wls}} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}(\mathbf{X}'\mathbf{W}\mathbf{y}) \tag{4}$$

where $\mathbf{W} = \text{diag}(\mathbf{w})$. These coefficients minimize the weighted sum of the squared residuals; that is, $\widehat{\boldsymbol{\beta}}_{\text{wls}} = \text{argmin}_{\boldsymbol{\beta}} \sum_{i=1}^{N} w_i(y_i - \mathbf{x}_i\boldsymbol{\beta})^2$. When all weights are equal to 1—that is, $\mathbf{w} = \mathbf{1}_N$—then $\widehat{\boldsymbol{\beta}}_{\text{wls}} = \widehat{\boldsymbol{\beta}}_{\text{ols}}$.

The conventional homoskedastic estimate of variance for the weighted regression is

$$\widehat{\mathbf{V}}_{\text{hom}} = s^2_w(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1} \tag{5}$$

where $s^2_w = (\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}_{\text{wls}})'\mathbf{W}(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}_{\text{wls}})/(N-k)$.

The weight vector, $\mathbf{w}$, is a function of the $N \times 1$ weight vector specified by the user, $\mathbf{v}$, and the weight type requested for estimation. The weight type can be one of frequency weights (`fweights`), analytic weights (`aweights`), sampling weights (`pweights`), or importance weights (`iweights`).

When `aweights` or `pweights` are specified, the weights are first normalized to sum up to N, $\mathbf{w} = (N/\mathbf{1}_N{}'\mathbf{v})\mathbf{v}$, and then used in calculations. Furthermore, when `pweights` are specified, heteroskedasticity-robust standard errors, `vce(robust)`, are calculated by default.

When `fweights` are specified, the unnormalized weights $\mathbf{w} = \mathbf{v}$ are used instead. The sample size is now assumed to be equal to the sum of the weights; that is, $N = \mathbf{1}_N{}'\mathbf{w}$. Standard errors and degrees of freedom are calculated using this modified sample size.

`iweights`, for historical reasons, are treated like `fweights` when robust standard errors are not specified. On the other hand, when `vce(robust)`, `vce(cluster` *clustvarlist*`)`, `vce(hc2)`, or `vce(hc3)` is specified, `iweights` are treated like `aweights`.

Heteroskedasticity-robust and cluster–robust variance estimation

General notation

We first establish a general notation for consistent variance estimation in the presence of heteroskedastic errors or one-way clustering. For multiway clustering, see *Multiway clustering* below.

Let G be the total number of clusters defined in the cluster variable *clustvar*, N_g be the number of observations in cluster g, and $N = \sum_{g=1}^{G} N_g$ be the total number of observations in the sample. Let w_{gi} be the weight of observation i in cluster g. Define the weighting matrices $\mathbf{W} = \text{diag}(w_{11}, w_{12}, \ldots, w_{GN_G})$ and $\mathbf{W}_g = \text{diag}(w_{g1}, \ldots, w_{gN_g})$. In the absence of weights, $\mathbf{W} = \mathbf{I}_N$ and $\mathbf{W}_g = \mathbf{I}_{N_g}$.

For cluster g, let $\widetilde{\boldsymbol{\epsilon}}_g = \mathbf{W}_g^{1/2}(\mathbf{y}_g - \mathbf{X}_g\widehat{\boldsymbol{\beta}})$ be the vector of weighted model residuals and $\widetilde{\mathbf{X}}_g = \mathbf{W}_g^{1/2}\mathbf{X}_g$ be the matrix of weighted regressors. Similarly, let $\widetilde{\mathbf{X}} = \mathbf{W}^{1/2}\mathbf{X}$ be the matrix of weighted regressors for the whole estimation sample.

vce(cluster clustvar)

The cluster–robust variance estimator of Arellano (1987) and Liang and Zeger (1986) is defined as

$$\widehat{\mathbf{V}}_{\text{cluster}} = \left(\frac{G}{G-1}\right)\left(\frac{N-1}{N-k}\right)(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\left(\sum_{g=1}^{G}\widetilde{\mathbf{X}}_g'\widetilde{\boldsymbol{\epsilon}}_g\widetilde{\boldsymbol{\epsilon}}_g'\widetilde{\mathbf{X}}_g\right)(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1} \tag{6}$$

Hansen (2007) showed that the term $G/(G-1)$ improves performance in Monte Carlo simulations when there are few equally sized clusters. The term $(N-1)/(N-k)$ was introduced by Fuller et al. (1986) so that the degrees-of-freedom correction in (6) nests the degrees-of-freedom corrections used in $\widehat{\mathbf{V}}_{\text{robust}}$ and $\widehat{\mathbf{V}}_{\text{hom}}$ when there are no clusters (and therefore $G = N$).

vce(robust)

In the absence of clusters and when `fweights` are specified, the traditional heteroskedasticity-robust variance estimator is

$$\widehat{\mathbf{V}}_{\text{robust}} = \frac{N}{(N-k)}(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\left(\sum_{i=1}^{N}\widetilde{\epsilon}_i^2\widetilde{\mathbf{X}}_i'\widetilde{\mathbf{X}}_i\right)(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1} \tag{7}$$

When `aweights`, `pweights`, or `iweights` are specified instead, the traditional heteroskedasticity-robust variance estimator is given by

$$\widehat{\mathbf{V}}_{\text{robust}} = \frac{N}{(N-k)}(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\left(\sum_{i=1}^{N}w_i^2\epsilon_i^2\widetilde{\mathbf{X}}_i'\widetilde{\mathbf{X}}_i\right)(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1} \tag{8}$$

vce(hc2 clustvar)

When `vce(hc2` *clustvar*`)` is specified, the variance estimator is

$$\widehat{\mathbf{V}}_{\text{hc2 cluster}} = (\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\left(\sum_{g=1}^{G}\widetilde{\mathbf{X}}_g'\mathbf{M}_g^{+1/2}\widetilde{\boldsymbol{\epsilon}}_g\widetilde{\boldsymbol{\epsilon}}_g'\mathbf{M}_g^{+1/2}\widetilde{\mathbf{X}}_g\right)(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1} \tag{9}$$

where $\mathbf{M}_g^{+1/2} = \{\mathbf{I}_{n_g} - \widetilde{\mathbf{X}}_g(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\widetilde{\mathbf{X}}_g'\}^{+1/2}$ is the Moore–Penrose generalized inverse of the symmetric square root of the partial annihilator matrix.

vce(hc2)

In the absence of clusters and when all partial annihilator matrices $\mathbf{M}_g$ are invertible, the heteroskedasticity-robust variance estimator produced when `vce(hc2)` is specified is given by

$$\widehat{\mathbf{V}}_{\text{hc2}} = (\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\left\{\sum_{i=1}^{N}(\tau_i - \tilde{h}_{ii})^{-1}\tilde{\epsilon}_i^2\widetilde{\mathbf{X}}_i'\widetilde{\mathbf{X}}_i\right\}(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1} \tag{10}$$

where $\tilde{h}_{ii} = \tilde{\mathbf{x}}_i(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\tilde{\mathbf{x}}_i'$ is the leverage value of observation i and τ_i equals w_i if using `fweights` or 1 otherwise. Invertibility of all partial annihilation matrices $\mathbf{M}_g$ implies that $\tilde{h}_{ii} < 1$ for all i.

If invertibility fails for some observations, then these observations are excluded from the summation in (10) when calculating the VCE. This behavior is consistent with (9) and the definition of the Moore–Penrose generalized inverse, because $(0)^{+1/2} = 0$.

vce(hc3 clustvar) (StataNow)

When `vce(hc3` *clustvar*`)` is specified, the variance estimator is

$$\widehat{\mathbf{V}}_{\text{hc3 cluster}} = (\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\left(\sum_{g=1}^{G}\widetilde{\mathbf{X}}_g'\mathbf{M}_g^{+}\tilde{\boldsymbol{\epsilon}}_g\tilde{\boldsymbol{\epsilon}}_g'\mathbf{M}_g^{+}\widetilde{\mathbf{X}}_g\right)(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1} \tag{11}$$

where $\mathbf{M}_g^{+} = \{\mathbf{I}_{n_g} - \widetilde{\mathbf{X}}_g(\widetilde{\mathbf{X}}\widetilde{\mathbf{X}}')^{-1}\widetilde{\mathbf{X}}_g'\}^{+}$ is the Moore–Penrose generalized inverse of the partial annihilator matrix.

When `vce(hc3` *clustvar*`)` is specified with suboption `hansen`, the variance estimator is computed as the jackknife estimator of Hansen (2025b). In the typical case where $\mathbf{M}_g$ is invertible for each cluster, this estimator is identical to $\widehat{\mathbf{V}}_{\text{hc3 clustvar}}$.

vce(hc3)

In the absence of clusters and when all partial annihilator matrices $\mathbf{M}_g$ are invertible, the heteroskedasticity-robust variance estimator produced when `vce(hc3)` is specified is given by

$$\widehat{\mathbf{V}}_{\text{hc3}} = (\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\left\{\sum_{i=1}^{N}\tau_i(\tau_i - \tilde{h}_{ii})^{-2}\tilde{\epsilon}_i^2\widetilde{\mathbf{X}}_i'\widetilde{\mathbf{X}}_i\right\}(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1} \tag{12}$$

where $\tilde{h}_{ii} = \tilde{\mathbf{x}}_i(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\tilde{\mathbf{x}}_i'$ is the leverage value of observation i and τ_i equals w_i if using `fweights` or 1 otherwise. Invertibility of all partial annihilation matrices $\mathbf{M}_g$ implies that $\tilde{h}_{ii} < 1$ for all i.

If invertibility fails for some observations, then these observations are excluded from the summation in (12) when calculating the VCE. This behavior is consistent with (11) and the definition of the Moore–Penrose generalized inverse, because $0^{+} = 0$.

When `vce(hc3)` is specified with suboption `hansen`, the variance estimator is computed as the jackknife estimator of Hansen (2025b). In the typical case where $\tau_i - \tilde{h}_{ii}$ is nonzero for each observation, this estimator is identical to $\widehat{\mathbf{V}}_{\text{hc3}}$.

Multiway clustering

When `vce(cluster` *clustvarlist*`)` is specified with more than one variable, the variance–covariance estimator uses multiway cluster–robust variance estimation. This estimator is appropriate when we assume in our model that the errors between two observations may be correlated if the observations share a category in any of the clustering variables. For an introduction into the linear model with multiway clustering, see Cameron, Gelbach, and Miller (2011).

Suppose for simplicity that we have two clustering variables. Let $\widehat{\mathbf{V}}_{1,0}$ be the cluster–robust variance estimator (6) based on the clusters defined by the categories of the first clustering variable. Let $\widehat{\mathbf{V}}_{0,1}$ be defined analogously for the second clustering variable. Let $\widehat{\mathbf{V}}_{1,1}$ be the cluster–robust variance estimator based on the clusters defined by the intersection of both clustering variables; that is, assuming that two observations belong to the same cluster if and only if they share the same category in both of the clustering variables.

The two-way cluster–robust VCE is defined as

$$\widehat{\mathbf{V}}_{\text{two-way}} = \widehat{\mathbf{V}}_{1,0} + \widehat{\mathbf{V}}_{0,1} - \widehat{\mathbf{V}}_{1,1} \tag{13}$$

Now, suppose that there are p clustering variables. We first generalize the notation introduced for two-way clustering. For index $i \in \{0,1\}^p$, let $\widehat{V}_i$ denote the cluster–robust variance estimator (6) based on the clusters defined by the intersection of all the clustering variables for which their corresponding coordinate in i is a 1; that is, assuming that two observations belong to the same cluster if and only if they share the same category in all the clustering variables for which their corresponding coordinate in i is a 1.

Let $S_j \subset \{0,1\}^p$ denote the subset of indices for which $||i||_1 = j$. These are the indices that involve exactly j clustering variables. The multiway cluster–robust variance estimator is given by

$$\widehat{\mathbf{V}}_{\text{mway}} = \sum_{j=1}^{p} (-1)^{j-1} \sum_{i \in S_j} \widehat{\mathbf{V}}_i \tag{14}$$

In general, matrix $\widehat{\mathbf{V}}_{\text{mway}}$ is not guaranteed to be positive semidefinite. When this condition fails, we use the eigendecomposition of $\widehat{\mathbf{V}}_{\text{mway}}$ to obtain a positive semidefinite estimator of the variance–covariance matrix. Let $\boldsymbol{\lambda}$ be the vector containing the eigenvalues of $\widehat{\mathbf{V}}_{\text{mway}}$, and let $\mathbf{Q}$ be the matrix containing the corresponding eigenvectors (as columns). When $\widehat{\mathbf{V}}_{\text{mway}}$ is not positive semidefinite, vector $\boldsymbol{\lambda}$ will contain negative coordinates. Let $\boldsymbol{\lambda}^+$ contain all the nonnegative elements of $\boldsymbol{\lambda}$ and a zero whenever the corresponding coordinate in $\boldsymbol{\lambda}$ is negative. Then define the variance–covariance matrix estimator as $\widehat{\mathbf{V}}^+_{\text{mway}} = \mathbf{Q}\,\text{diag}(\boldsymbol{\lambda}^+)\mathbf{Q}'$. This estimator is guaranteed to be positive semidefinite.

Degrees of freedom

For homoskedastic and heteroskedasticity-robust variance estimators, p-values and confidence intervals are derived from a t distribution with $N - k$ degrees of freedom, where k is the number of right-hand-side variables (including the constant).

For one-way cluster–robust variance estimators, p-values and confidence intervals are derived from a t distribution with $G - 1$ degrees of freedom, where G is the total number of clusters. For multiway clustering with p clustering variables, the degrees of freedom equals $\min\{G_1, G_2, \ldots, G_p\} - 1$, where G_q is the total number of clusters defined by the qth clustering variable.

Adjusted degrees of freedom

`vce(hc2 [`*clustvar*`], dfadjust)` directs `regress` to compute the adjusted degrees of freedom described by Imbens and Kolesár (2016). To simplify notation, let $\widetilde{\mathbf{H}} = \widetilde{\mathbf{X}}(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\widetilde{\mathbf{X}}'$, $\widetilde{\mathbf{H}}_{gg} = \widetilde{\mathbf{X}}_g(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\widetilde{\mathbf{X}}_g'$, and $\widetilde{\mathbf{H}}_g = \widetilde{\mathbf{X}}(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\widetilde{\mathbf{X}}_g'$.

For the lth regressor, define the $N \times G$ matrix $\mathbf{G}_{(l)}$ such that its gth column is

$$(\mathbf{I}_N - \widetilde{\mathbf{H}})_g{}'(\mathbf{I}_{N_g} - \widetilde{\mathbf{H}}_{gg})^{+1/2}\widetilde{\mathbf{X}}_g(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\mathbf{e}_l$$

where $\mathbf{e}_l$ is a $k \times 1$ unitary vector with a 1 in the lth coordinate and $(\mathbf{I}_N - \widetilde{\mathbf{H}})_g$ is the $N \times N_g$ matrix consisting of the N_g columns of the $N \times N$ matrix $(\mathbf{I}_N - \widetilde{\mathbf{H}})$ corresponding to cluster g. Bell and McCaffrey's (2002) adjusted degrees of freedom for the lth regressor is

$$K_l = \frac{\mathrm{tr}(\mathbf{G}_{(l)}'\mathbf{G}_{(l)})^2}{\mathrm{tr}(\mathbf{G}_{(l)}'\mathbf{G}_{(l)})^2} = \frac{(\sum_{i=1}^{G}\lambda_i)^2}{\sum_{i=1}^{G}\lambda_i^2} \tag{15}$$

where tr is the trace function and λ_i are the eigenvalues of matrix $\mathbf{G}_{(l)}{}'\mathbf{G}_{(l)}$.

The calculation of the eigenvalues can take a long time when G is large or when there are no clusters (and therefore $G = N$). Following Kolesár (2023), we can speed the calculation by defining

$$\mathbf{a}_g = (\mathbf{I}_{N_g} - \widetilde{\mathbf{H}}_{gg})^{+1/2}\widetilde{\mathbf{X}}_g(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\mathbf{e}_l$$

$$\mathbf{b}_g = \widetilde{\mathbf{H}}_g\mathbf{a}_g$$

$$\mathbf{A} = (\mathbf{a}_1'\mathbf{a}_1, \ldots, \mathbf{a}_G'\mathbf{a}_G)' = (\alpha_1, \alpha_2, \ldots, \alpha_G)'$$

$$\mathbf{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_G) = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1G} \\ \vdots & \vdots & \ddots & \vdots \\ b_{N1} & b_{N2} & \cdots & b_{NG} \end{pmatrix}$$

Then $\mathbf{G}'\mathbf{G} = \mathrm{diag}(\mathbf{A}) - \mathbf{B}'\mathbf{B}$.

When clustered standard errors are requested, the adjusted degrees of freedom for the lth regressor are given by

$$K_l = \frac{(\sum_{g=1}^{G}\alpha_g - \sum_{i=1}^{N}\sum_{g=1}^{G}b_{ig}^2)^2}{\sum_{g=1}^{G}\alpha_g^2 - 2\sum_{g=1}^{G}\alpha_g\sum_{i=1}^{N}b_{ig}^2 + \sum_{g=1}^{G}\sum_{j=1}^{G}(\mathbf{b}_g{}'\mathbf{b}_j)^2}$$

Note that, in the clustered case, weights only enter the adjusted degrees of freedom through the weighted matrix of regressors $\widetilde{\mathbf{X}}$.

The degrees of freedom can be computed efficiently using a QR decomposition. For example, by decomposing $\mathbf{X} = \mathbf{QR}$, where $\mathbf{Q}$ is an $N \times k$ orthonormal matrix and $\mathbf{R}$ is a $k \times k$ upper triangular, we can rewrite the matrix $\mathbf{B}$ so that it has dimension $k \times G$ instead of $N \times G$.

When there are no clusters, the adjusted degrees of freedom is given by

$$K_l = \frac{(\sum_{i=1}^{N}w_i\alpha_i - \sum_{i=1}^{k}\sum_{j=1}^{N}b_{ij}^2)^2}{\sum_{i=1}^{N}w_i\alpha_i^2 - 2\sum_{j=1}^{N}\alpha_j\sum_{i=1}^{k}b_{ij}^2 + \sum_{i=1}^{N}\sum_{j=1}^{N}(\mathbf{b}_i{}'\mathbf{b}_j)^2}$$

Here we substituted k for N in the row dimension of $\mathbf{B}$ as it is when using QR decomposition to perform the computations.

`vce(hc3 [`*clustvar*`], dfadjust)` also directs `regress` to compute the adjusted degrees of freedom described by Imbens and Kolesár (2016). The adjusted degrees of freedom is computed identically as for `vce(hc2 [`*clustvar*`], dfadjust)`, except that $(\mathbf{I}_{N_g} - \widetilde{\mathbf{H}}_{gg})^{+1/2}$ is substituted by $(\mathbf{I}_{N_g} - \widetilde{\mathbf{H}}_{gg})^{+}$ in the definitions of matrix $\mathbf{G}_{(l)}$ and of the $\mathbf{a}_g$ vectors.

`vce(hc3 [`*clustvar*`], hansen)` computes the degrees of freedom in Hansen (2025b). In the typical case of cluster invertibility, these are identical to the degrees of freedom computed using `vce(hc3 [`*clustvar*`], dfadjust)`. The scaling factors a_l from Hansen (2025b) are also computed.

With cluster invertibility, the scaling factor for the lth regressor can be written using the above notation as

$$a_l = \sqrt{\frac{\left(\sum_{i=1}^{N} w_i\alpha_i - \sum_{i=1}^{k}\sum_{j=1}^{N} b_{ij}^2\right)^2}{\mathbf{e}_l'(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\mathbf{e}_l}}$$

These scaling factors are used to adjust confidence intervals and p-values. For example, the 95% confidence interval for the coefficient β_l is now

$$\mathrm{CI}_{\beta_l} = \widehat{\beta}_l \pm \frac{t_{K_l}^{97.5}}{a_l}\widehat{\mathrm{SE}}_l$$

HAC standard errors (StataNow)

When `vce(hac` *kernel* `[#])` is specified, then

$$\widehat{\mathbf{V}}_{\mathrm{HAC}} = \frac{N}{N-k}(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}\left\{\sum_j \tilde{\epsilon}_j^2\tilde{\mathbf{x}}_j\tilde{\mathbf{x}}_j' + \sum_{l=1}^{l=N-1}\sum_{j=l+1}^{j=N} K(l,m)\tilde{\epsilon}_j\tilde{\epsilon}_{j-l}\left(\tilde{\mathbf{x}}_j\tilde{\mathbf{x}}_{j-l}' + \tilde{\mathbf{x}}_{j-l}\tilde{\mathbf{x}}_j'\right)\right\}(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}})^{-1}$$

where $\tilde{\mathbf{x}}_i = \sqrt{w_i}\mathbf{x}_i$ and $K(\cdot)$ is a real-valued kernel function. The maximum number of lags, m, equals $N-2$ by default or # if `vce(hac` *kernel* #`)` is specified. Define $z = l/(m+1)$. If *kernel* is `nwest`, then

$$K(l,m) = \begin{cases} 1-z & 0 \le z \le 1 \\ 0 & \text{otherwise} \end{cases}$$

If *kernel* is `gallant`, then

$$K(l,m) = \begin{cases} 1 - 6z^2 + 6z^3 & 0 \le z \le 0.5 \\ 2(1-z)^3 & 0.5 < z \le 1 \\ 0 & \text{otherwise} \end{cases}$$

If *kernel* is `quadraticspectral`, then

$$K(l,m) = \begin{cases} 1 & z = 0 \\ 3\left\{\sin(\theta)/\theta - \cos(\theta)\right\}/\theta^2 & \text{otherwise} \end{cases}$$

where $\theta = 6\pi z/5$.

If vce(hac *kernel* opt) is specified, then regress uses Newey and West's (1994) automatic lag-selection algorithm, which proceeds as follows. Define **h** to be a $k \times 1$ vector containing ones in all rows except for the row corresponding to the constant term (if present); that row contains a zero. Define

$$f_j = \tilde{\epsilon}_j \mathbf{h}' \tilde{\mathbf{x}}_j$$

$$\hat{\sigma}_k = \frac{1}{N} \sum_{i=k+1}^{N} f_j f_{j-k} \qquad k = 0, \ldots, m^*$$

$$\hat{s}^{(q)} = 2 \sum_{k=1}^{m^*} \hat{\sigma}_j j^q$$

$$\hat{s}^{(0)} = \hat{\sigma}_0 + 2 \sum_{k=1}^{m^*} \hat{\sigma}_j$$

$$\hat{\gamma} = c_\gamma \left\{ \left(\frac{\hat{s}^{(q)}}{\hat{s}^{(0)}} \right)^2 \right\}^{1/2q+1}$$

$$m = \hat{\gamma} N^{1/(2q+1)}$$

where q, m^*, and c_γ depend on the kernel specified:

Kernel	q	m^*	c_γ
Bartlett	1	int $\{20(T/100)^{2/9}\}$	1.1447
Parzen	2	int $\{20(T/100)^{4/25}\}$	2.6614
Quadratic spectral	2	int $\{20(T/100)^{2/25}\}$	1.3221

int(x) denotes the integer obtained by truncating x toward zero. For the Bartlett and Parzen kernels, the optimal lag is min{int(m), m^*}. For the quadratic spectral, the optimal lag is min$\{m, m^*\}$.

If vce(hac *kernel* opt #) is specified, then regress uses # instead of 20 in the definition of m^* above to select the optimal lag.

Acknowledgments

The robust estimate of variance was first implemented in Stata by Mead Over of the Center for Global Development, Dean Jolliffe of the World Bank, and Andrew Foster of the Department of Economics at Brown University (Over, Jolliffe, and Foster 1996).

The history of regression is long and complicated: the books by Stigler (1986) and Hald (1998) are devoted largely to the story. Legendre published first on least squares in 1805. Gauss published later in 1809, but he had the idea earlier. Gauss, and especially Laplace, tied least squares to a normal errors assumption. The idea of the normal distribution can itself be traced back to De Moivre in 1733. Laplace discussed a variety of other estimation methods and error assumptions over his long career, while linear models long predate either innovation. Most of this work was linked to problems in astronomy and geodesy.

A second wave of ideas started when Galton used graphical and descriptive methods on data bearing on heredity to develop what he called regression. His term reflects the common phenomenon that characteristics of offspring are positively correlated with those of parents but with regression slope such that offspring "regress toward the mean". Galton's work was rather intuitive: contributions from Pearson, Edgeworth, Yule, and others introduced more formal machinery, developed related ideas on correlation, and extended application into the biological and social sciences. So most of the elements of regression as we know it were in place by 1900.

Pierre-Simon Laplace (1749–1827) was born in Normandy and was early recognized as a remarkable mathematician. He weathered a changing political climate well enough to rise to Minister of the Interior under Napoleon in 1799 (although only for 6 weeks) and to be made a Marquis by Louis XVIII in 1817. He made many contributions to mathematics and physics, his two main interests being theoretical astronomy and probability theory (including statistics). Laplace transforms are named for him.

Adrien-Marie Legendre (1752–1833) was born in Paris (or possibly in Toulouse) and educated in mathematics and physics. He worked in number theory, geometry, differential equations, calculus, function theory, applied mathematics, and geodesy. The Legendre polynomials are named for him. His main contribution to statistics is as one of the discoverers of least squares. He died in poverty, having refused to bow to political pressures.

Johann Carl Friedrich Gauss (1777–1855) was born in Braunschweig (Brunswick), now in Germany. He studied there and at Göttingen. His doctoral dissertation at the University of Helmstedt was a discussion of the fundamental theorem of algebra. He made many fundamental contributions to geometry, number theory, algebra, real analysis, differential equations, numerical analysis, statistics, astronomy, optics, geodesy, mechanics, and magnetism. An outstanding genius, Gauss worked mostly in isolation in Göttingen.

Francis Galton (1822–1911) was born in Birmingham, England, into a well-to-do family with many connections: he and Charles Darwin were first cousins. After an unsuccessful foray into medicine, he became independently wealthy at the death of his father. Galton traveled widely in Europe, the Middle East, and Africa, and became celebrated as an explorer and geographer. His pioneering work on weather maps helped in the identification of anticyclones, which he named. From about 1865, most of his work was centered on quantitative problems in biology, anthropology, and psychology. In a sense, Galton (re)invented regression, and he certainly named it. Galton also promoted the normal distribution, correlation approaches, and the use of median and selected quantiles as descriptive statistics. He was knighted in 1909.

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Also see

[R] **regress postestimation** — Postestimation tools for regress

[R] **regress postestimation diagnostic plots** — Postestimation plots for regress

[R] **regress postestimation time series** — Postestimation tools for regress with time series

[R] **anova** — Analysis of variance and covariance

[R] **contrast** — Contrasts and linear hypothesis tests after estimation

[R] **hetregress** — Heteroskedastic linear regression

[R] **wildbootstrap** — Wild cluster bootstrap inference

[BAYES] **bayes: regress** — Bayesian linear regression

[BMA] **bmaregress** — Bayesian model averaging for linear regression

[CAUSAL] **Causal inference commands** — Introduction to causal inference commands

[FMM] **fmm: regress** — Finite mixtures of linear regression models

[LASSO] **Lasso intro** — Introduction to lasso

[META] **meta regress** — Meta-analysis regression

[MI] **Estimation** — Estimation commands for use with mi estimate

[SEM] Example 6 — Linear regression

[SEM] **Intro 5** — Tour of models

[SP] **spregress** — Spatial autoregressive models

[SVY] **svy estimation** — Estimation commands for survey data

[TS] **forecast** — Econometric model forecasting

[TS] **mswitch** — Markov-switching regression models

[U] 20 Estimation and postestimation commands

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